GENX: Porewater Ammonium

July 2026 Samples

2026-10-19

Contents

0.1 Assessing standard Curves . . . 4
0.2 Update Standard Log . . . 4
0.3 Dilution Corrections - ensure the latest dilution is kept . . . 6
0.4 Performance Check . . . 6
0.5 Analyze the Check Standards . . . 7
0.6 Analyze Blanks . . . 8
0.7 Analyze Duplicates . . . 9
0.8 Spikes . . . 10
0.9 Matrix Effects . . . 11
0.10 Create Log for QAQC . . . 12
0.11 Merge samples with Metadata & check all samples are present . . . 15
0.12 Visualize Data . . . 16
0.13 Organize . . . 20

##Run Information

```
cat("Run Information:  ") #lets you know what section you're in
```

```
## Run Information:
```

```
library(here)
```

```
## here() starts at C:/Users/blakleybd/OneDrive - Smithsonian Institution/Documents/GitHub/GCREW
```

```
#set the run date & user name
  sample_year <- "2026"
  sample_month <- "July"
  user <- "Brenden Blakley"

#identify the files you want to read in
  #read in as a list to accommodate multiple runs in a month
  NH3_files <- c("Raw Data/20260908_GENX_July2026.csv")

# Define the file path for QAQC log file - NO Need to change just check year
  file_path <- here("Porewater QAQC Logs", "GCReW_NH4_QAQC_Log.csv")
  file_path_QAQC <- here("Porewater QAQC Logs", "GCReW_SEAL_Check_QAQC_Log.csv")
  final_path <- "Processed Data/GCReW_GENX_Porewater_NH4_202607.csv"

#record any notes about the run or anything other info here:
  run_notes <- "CH1-80 and CH11-80 not collected"

  #Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

  cat(run_notes)
```

```
## CH1-80 and CH11-80 not collected
```

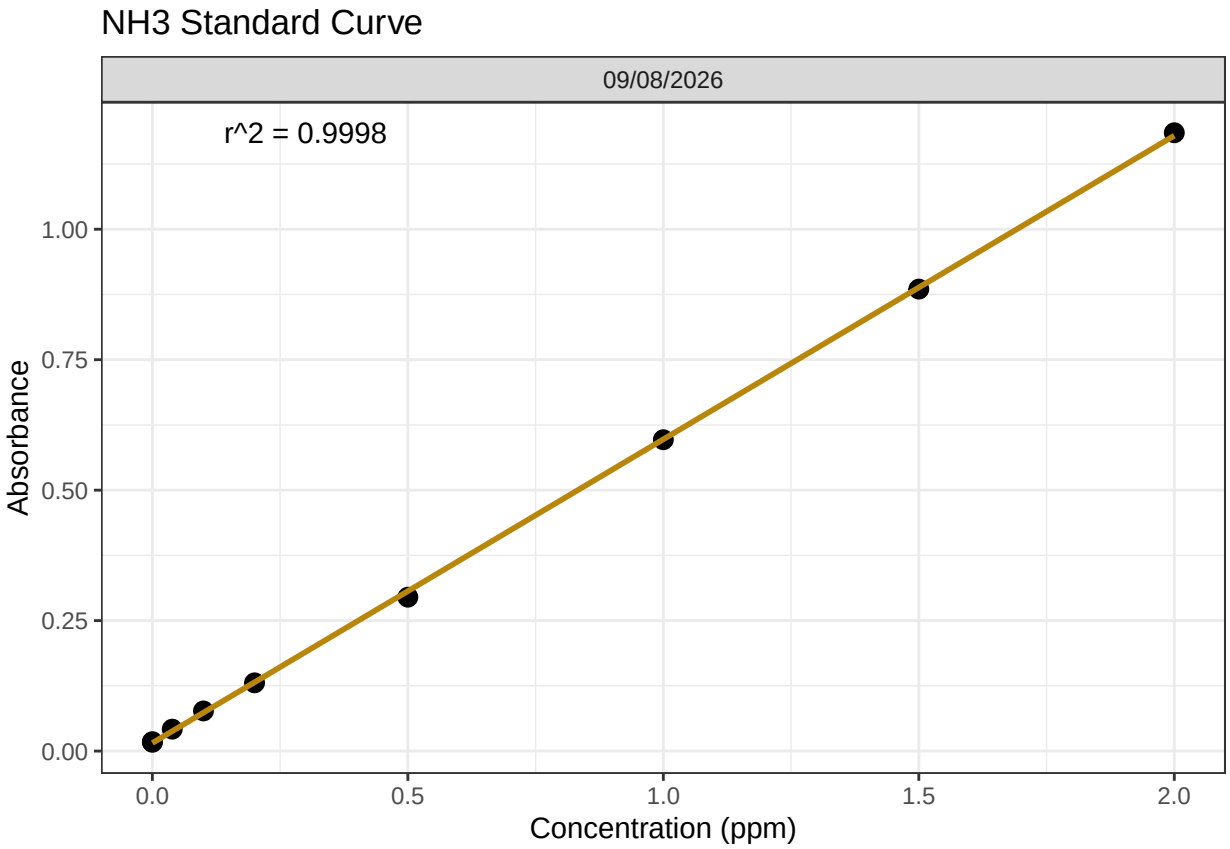
##Setup

##Read in metadata and create similar sample IDs for matching to samples

##Import Data & Clean

0.1 Assessing standard Curves

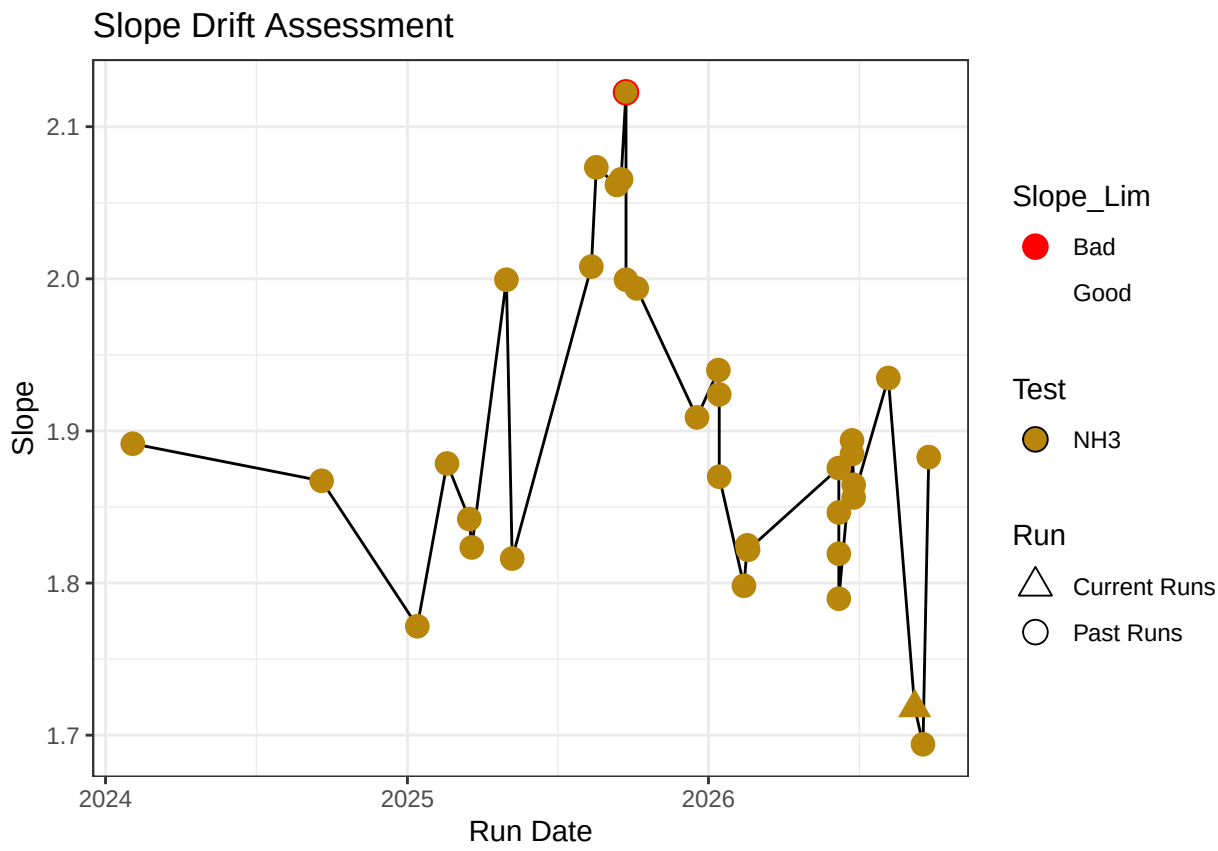
```
## `geom_smooth()` using formula = 'y ~ x'
```

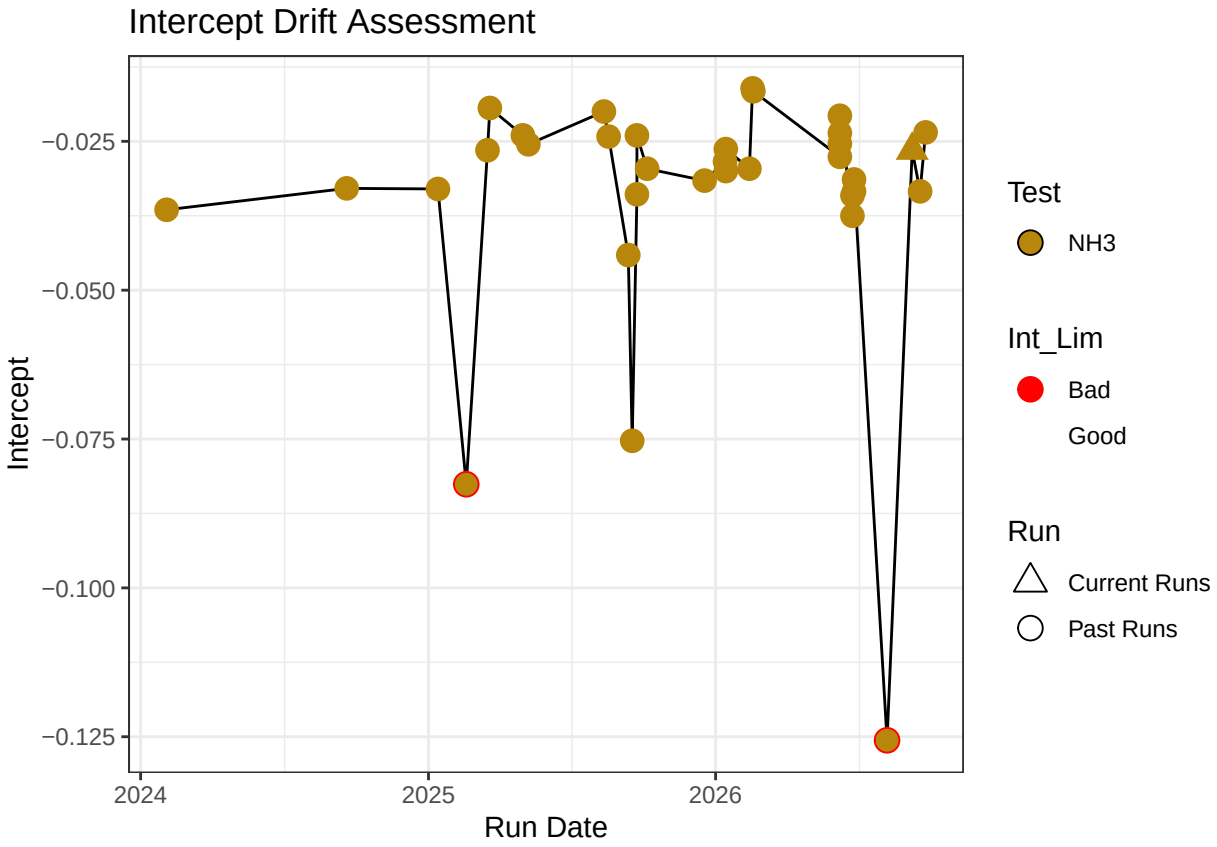


Test	Run_Date	R2_Flag	R2	Slope
Ammonia 2	09/08/2026	R2_Pass - PROCEED	0.9998395	1.718066

0.2 Update Standard Log

```
## `summarise()` has grouped output by 'Test', 'Intercept', 'Slope', 'R_Squared',
## 'User'. You can override using the `.groups` argument.
```


Slope Drift Assessment
Slope
2.1
2.0
1.9
1.8
1.7
2024
2025
2026
Run Date
Slope_Lim
Bad
Good
Test
NH3
Run
Current Runs
Past Runs



##Write Standard Log

0.3 Dilution Corrections - ensure the latest dilution is kept

```
## [1] "Reran Samples: c(\"CH3-80\", \"CH8-40\", \"CH8-80\", \"CH10-80\", \"CH12-40\", \"CH12-80\")"

## [1] "Diluted Samples: c(\"CH8-80\", \"CH10-80\", \"CH12-80\")"

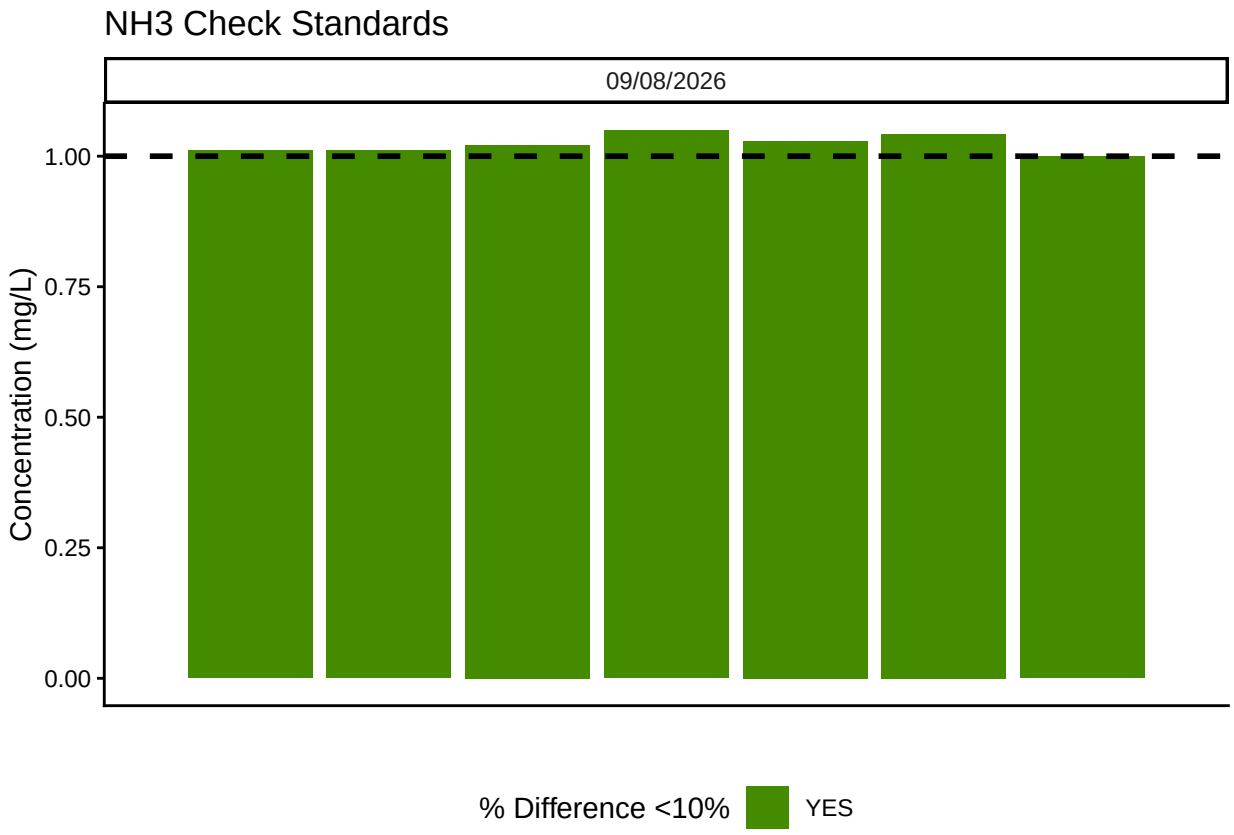
## [1] "Possible Naming Errors: c(\"CH3-80\", \"CH8-40\", \"CH12-40\")"
```

0.4 Performance Check

Test	Run_Date	PE_Flag	PE_Conc	PE_Target_Conc
Ammonia 2	09/08/2026	Performance Check Within 25% - PROCEED	1.218678	1.034

0.5 Analyze the Check Standards

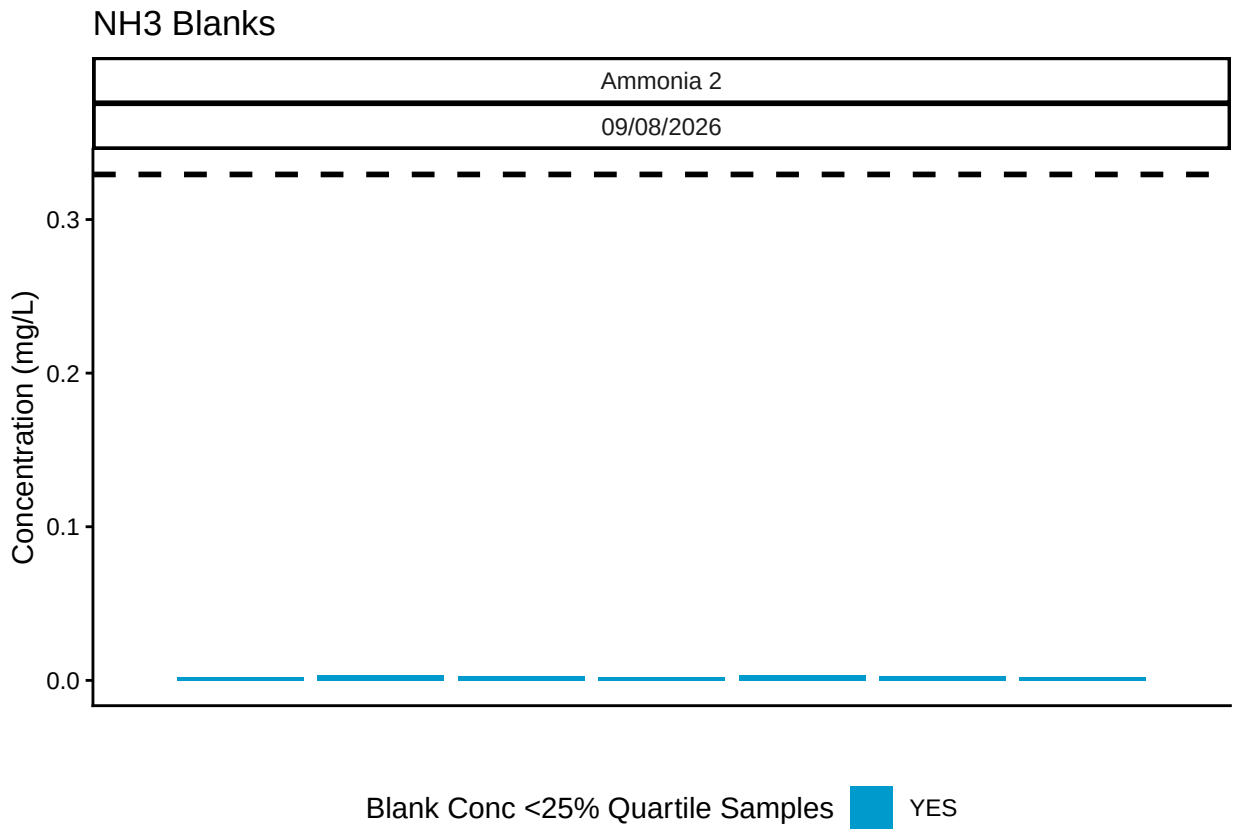
Test	Run_Date	RSV_Flag	RSV	RSV_Cutoff
Ammonia 2	09/08/2026	RSV WITHIN RANGE - PROCEED	0.0177087	0.25



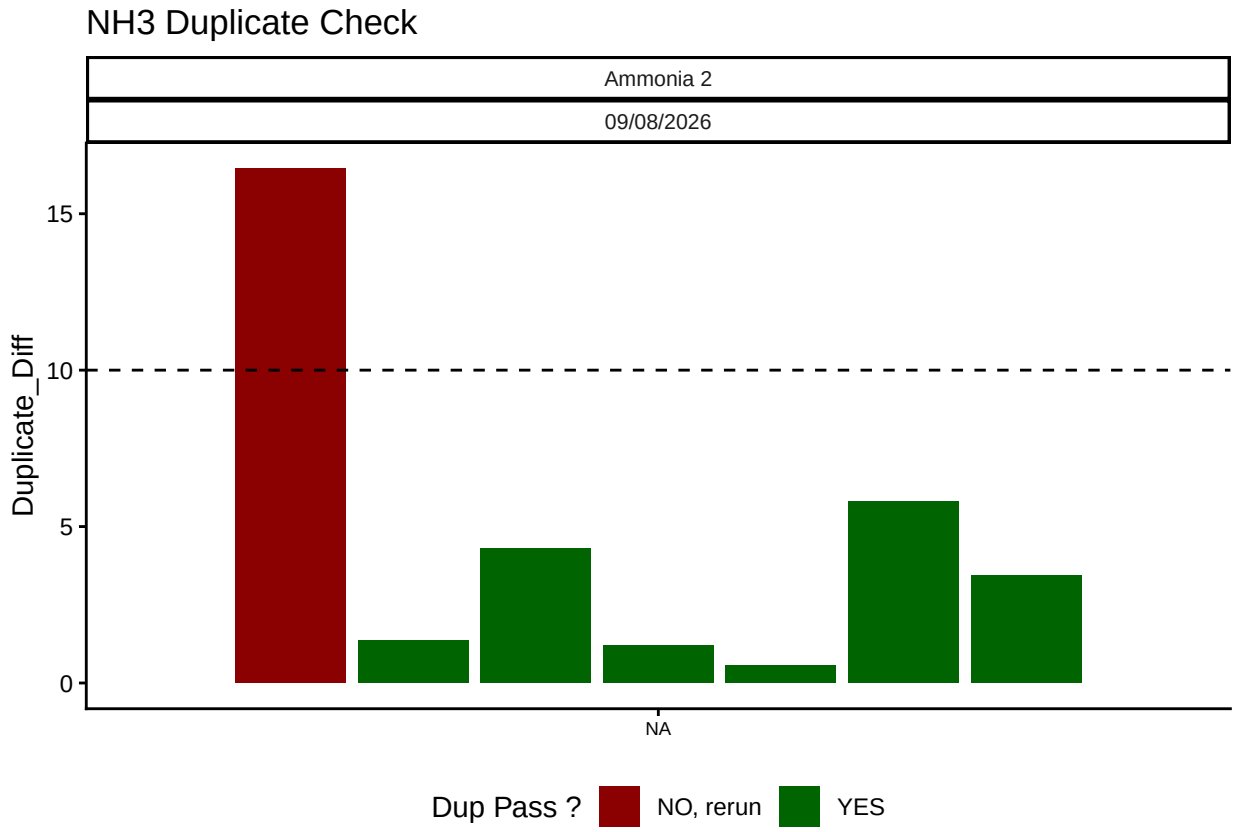
Test	Run_Date	CHK_Flag
Ammonia 2	09/08/2026	>60% of Checks Pass - PROCEED

0.6 Analyze Blanks

Test	Run_Date	BLK_Pct_Flag	Mean_Blk_Conc	Quantile_25
Ammonia 2	09/08/2026	>60% of Blanks Pass - PROCEED	0.0027779	0.3293365

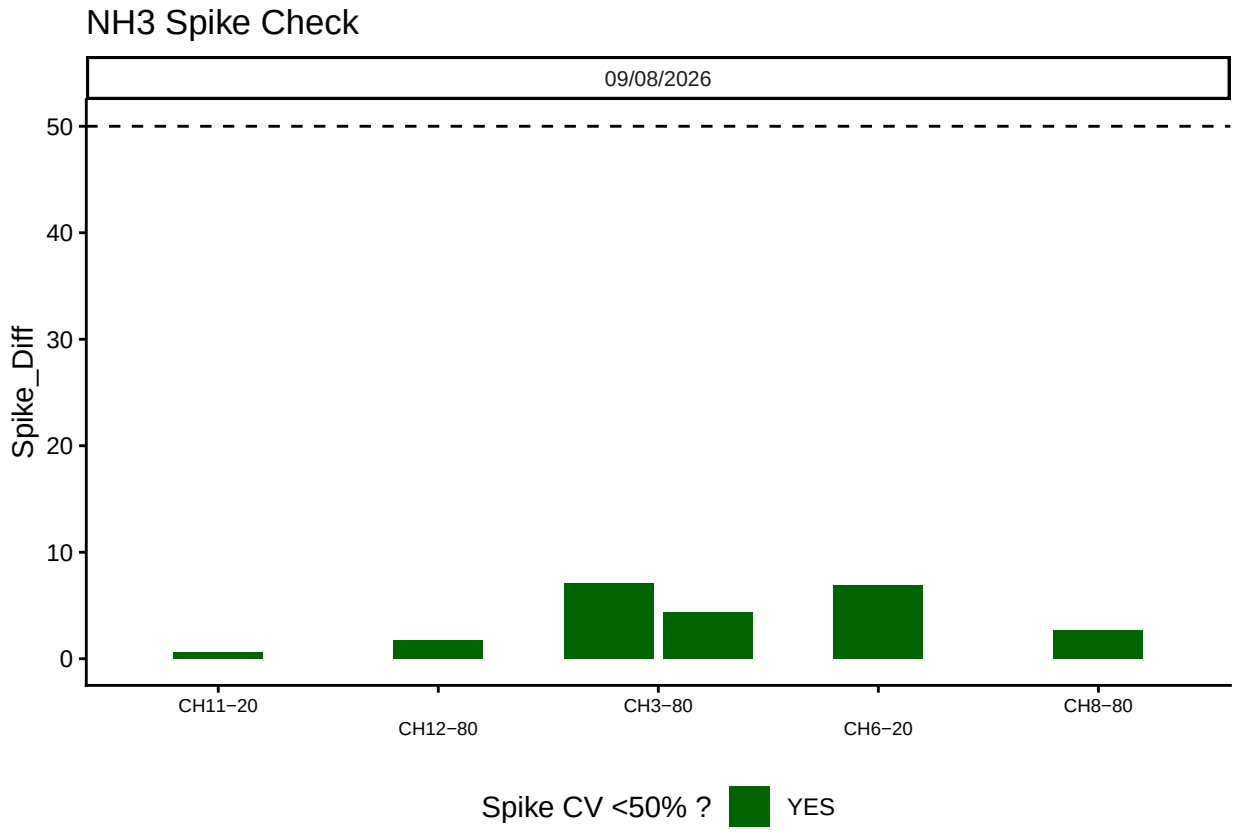


0.7 Analyze Duplicates



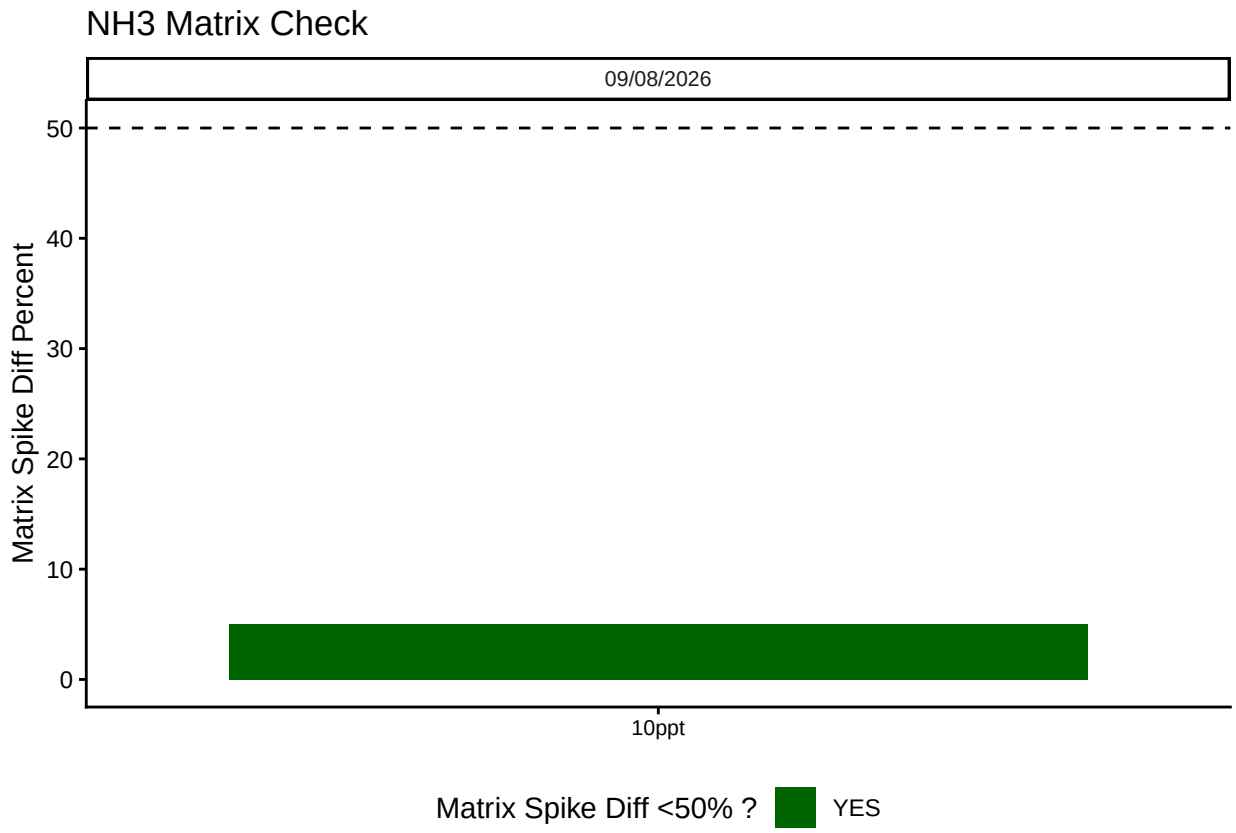
Test	Run_Date	Dup_Flags
Ammonia 2	09/08/2026	>60% of Dups Pass - PROCEED

0.8 Spikes



Test	Run_Date	Spike_Flags
Ammonia 2	09/08/2026	>60% of Spikes have a Diff <50% - PROCEED

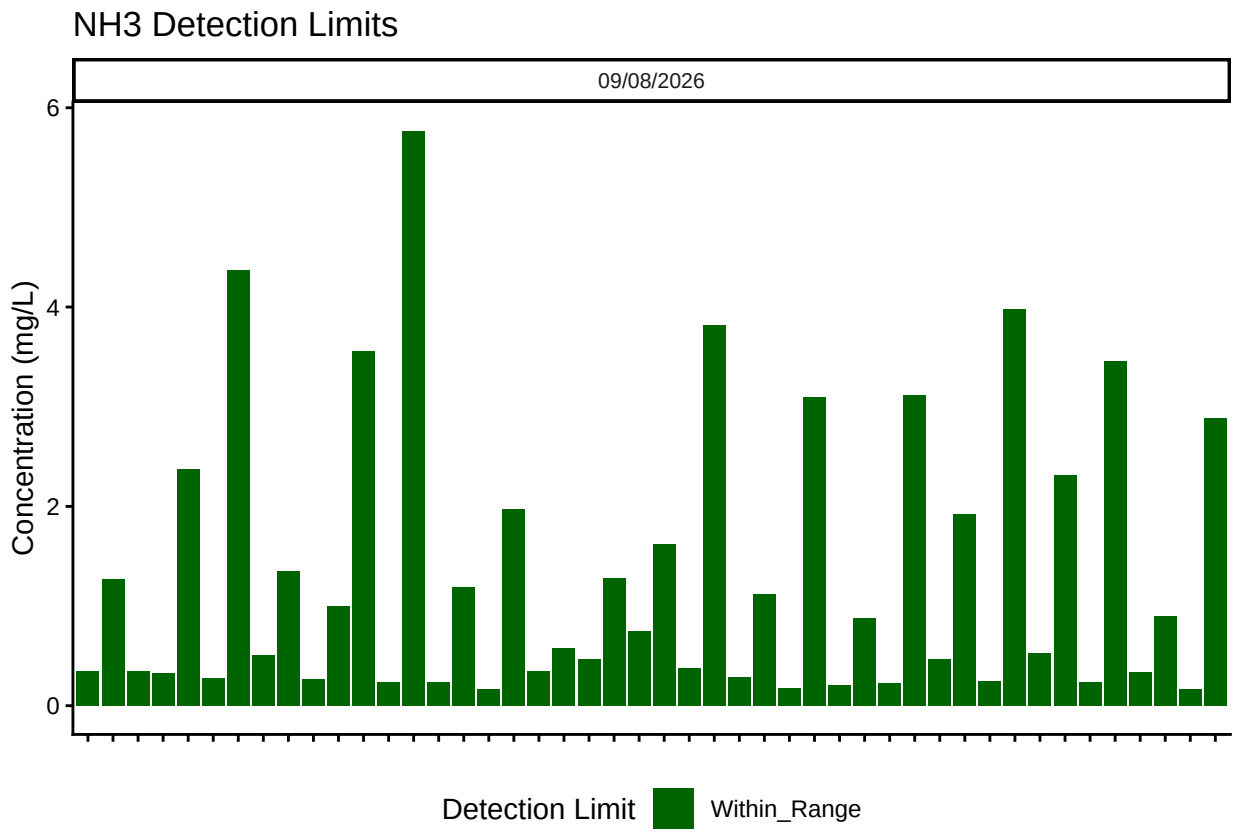
0.9 Matrix Effects



Test	Run_Date	Matrix_Flags
Ammonia 2	09/08/2026	Matrix Has Diff <50% - PROCEED

##Unit Conversion

##Sample Flagging - Within range of standard curve



##Add QAQC Flags

0.10 Create Log for QAQC

```
# Old_Log_QAQC <- tibble(Test = c(), Mean_PE_Conc= c(), Spike_CV= c(), Top_Std_Absorbance= c(), Run_Dat
Old_Log_QAQC <- read.csv(file_path_QAQC) %>% mutate(Run_Date = as.Date(as.character(Run_Date), tryForma

Top_Standards <- Slope_raw %>% group_by(Test, Run_Date) %>% filter(Sample_Name == "Standard 2.0000") %>

PE_Check <- NH3_peChk %>% group_by(Test, Run_Date) %>% dplyr::select(Test, Run_Date, Mean_Chk_Conc) %>%

Spikes_Log <- Spike_plot_df %>% group_by(Test, Run_Date) %>% dplyr::select(Test, Run_Date, Spike_Diff) %
                                                    Run_Date = Run_Date,
                                                    Spike_Percent_Diff = mean(Spike_Percent_Diff)) %>% u

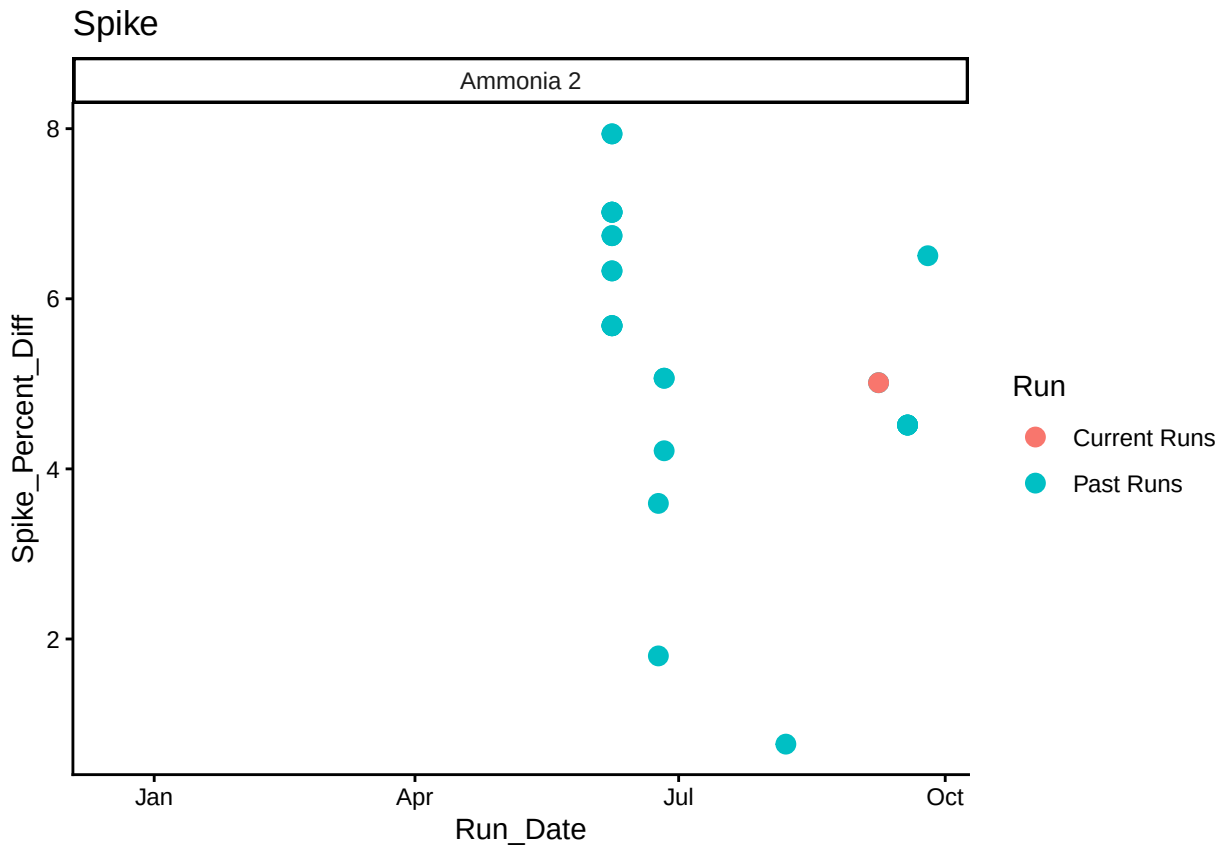
Joined_QAQC <- left_join(Top_Standards, PE_Check, by = join_by(Test, Run_Date)) %>%
  left_join(Spikes_Log, by = join_by(Test, Run_Date)) %>% mutate(Run = "Current Runs") %>% mutate(Run_Da
    group_by(dplyr::select(., -Run)) %>%
    summarize(Run = Run[max(1, which(Run != "Past Runs"))][1]) %>%
    ungroup()
```

```
## `summarise()` has grouped output by 'Test', 'Run_Date', 'Top_Std_Absorbance',
## 'Mean_PE_Conc'. You can override using the `.groups` argument.
```

```
Spike_Log_Plot <- ggplot(Joined_QAQC, aes(x = Run_Date, y = Spike_Percent_Diff)) +
  geom_point(size = 3, aes(col = Run)) +
 # geom_path(linewidth = 1) +
  facet_grid(~Test, scales = "free") +
  labs(title = "Spike") +
  theme_classic()

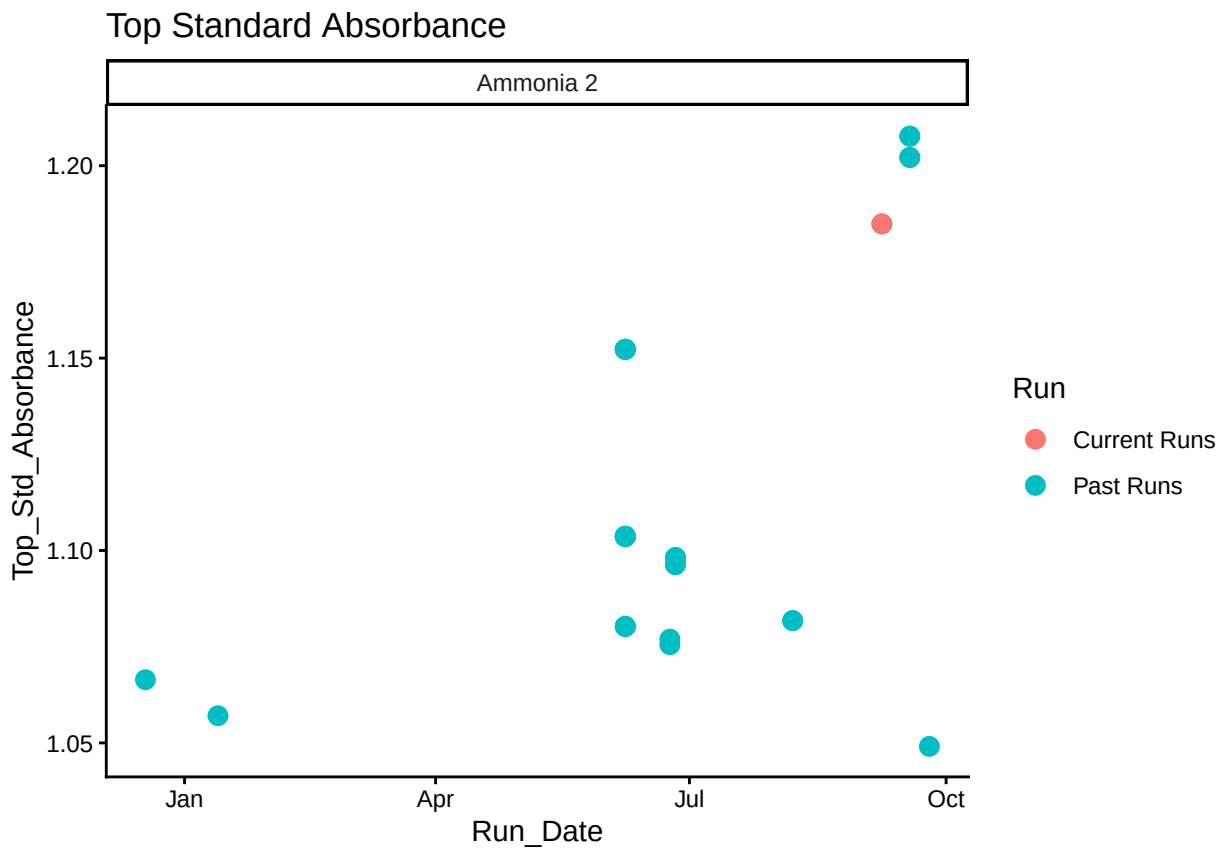
Spike_Log_Plot
```

```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
TopStd_Log_Plot <- ggplot(Joined_QAQC, aes(x = Run_Date, y = Top_Std_Absorbance)) +
  geom_point(size = 3, aes(col = Run)) +
 # geom_path(linewidth = 1) +
  facet_grid(~Test, scales = "free") +
  labs(title = "Top Standard Absorbance") +
  theme_classic()

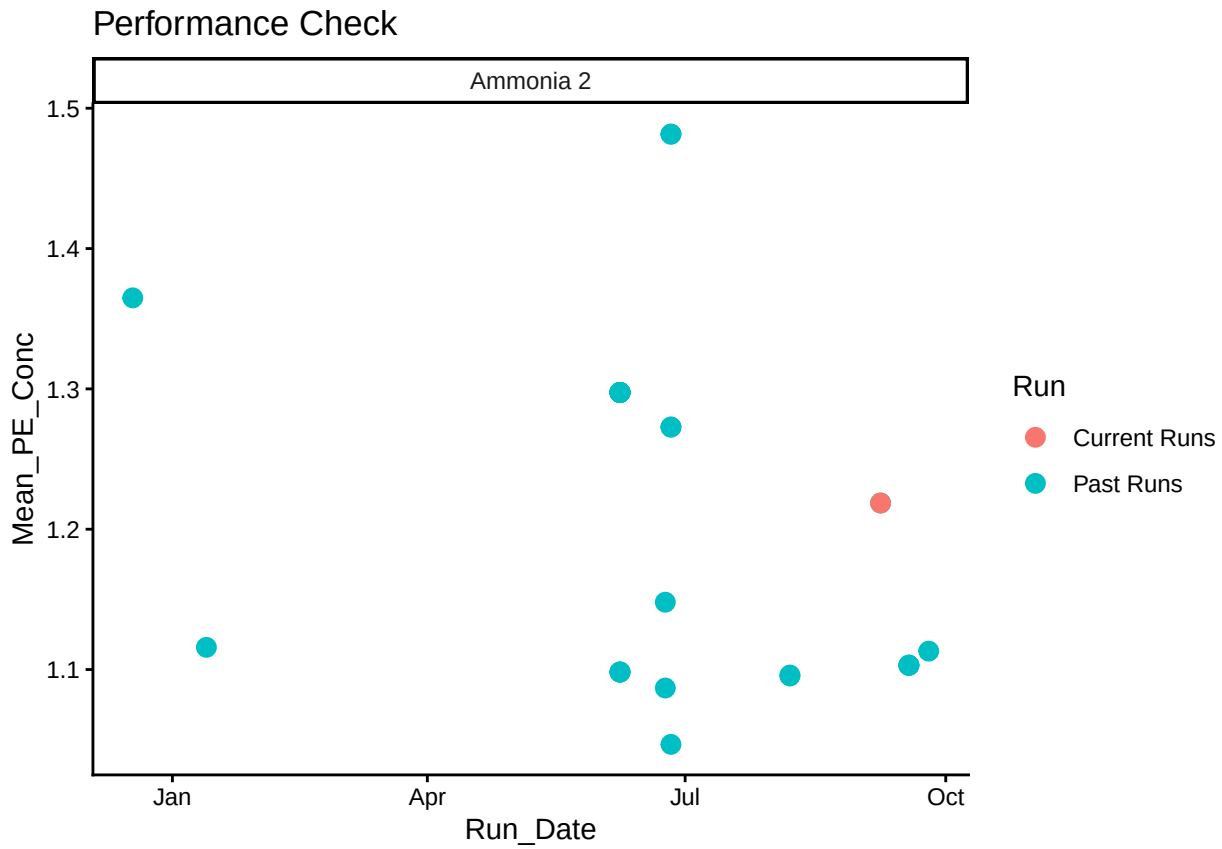
TopStd_Log_Plot
```




```
PE_Log_Plot <- ggplot(Joined_QAQC, aes(x = Run_Date, y = Mean_PE_Conc)) +
  geom_point(size = 3, aes(col = Run)) +
  #geom_path(linewidth = 1) +
  facet_grid(~Test, scales = "free") +
  labs(title = "Performance Check") +
  theme_classic()

PE_Log_Plot
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



##Write QAQC Log

0.11 Merge samples with Metadata & check all samples are present

```
## Merge Samples with Metadata

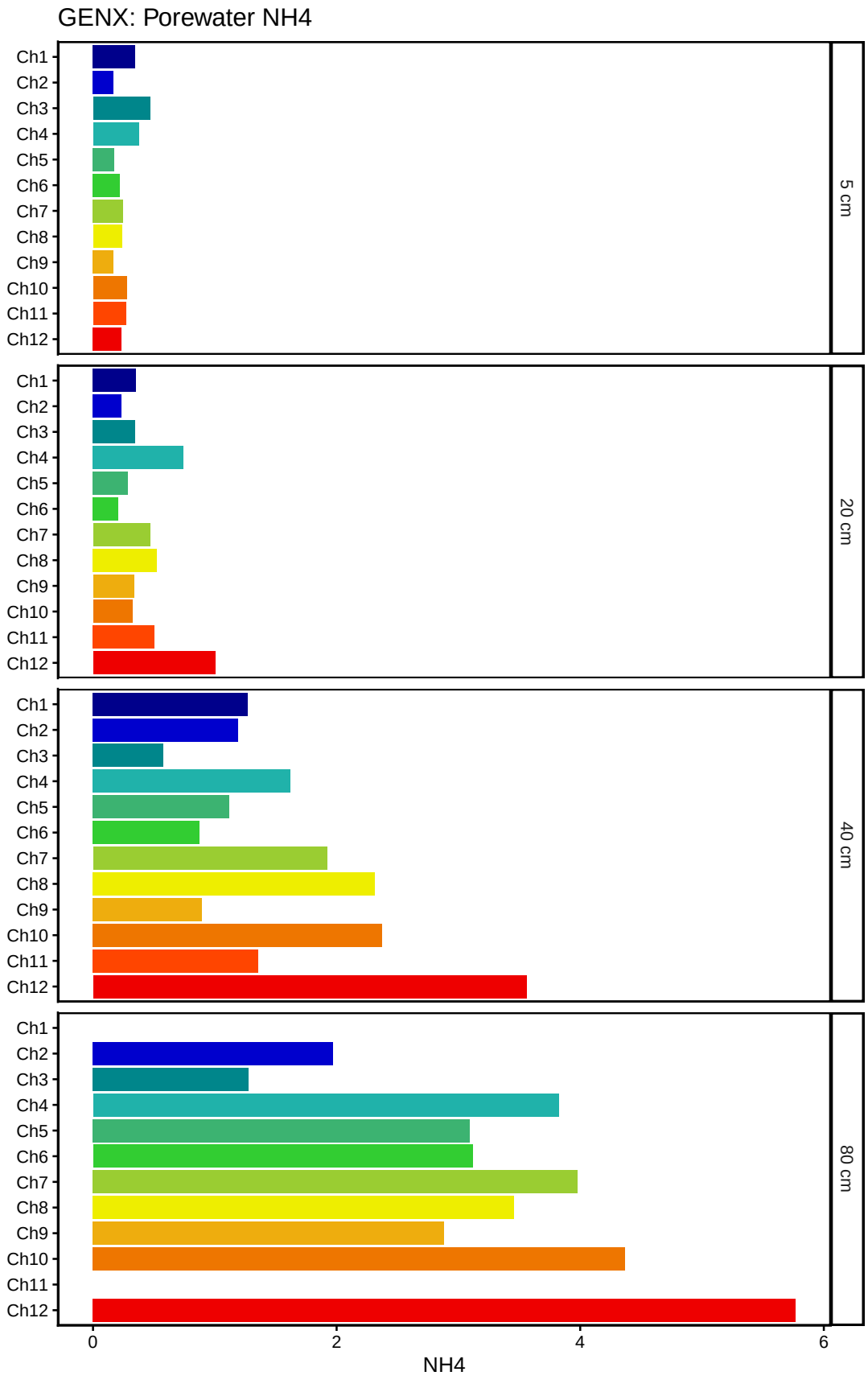
## Some sample IDs are missing from data.

## [1] "CH1-80"  "CH11-80"
```

0.12 Visualize Data

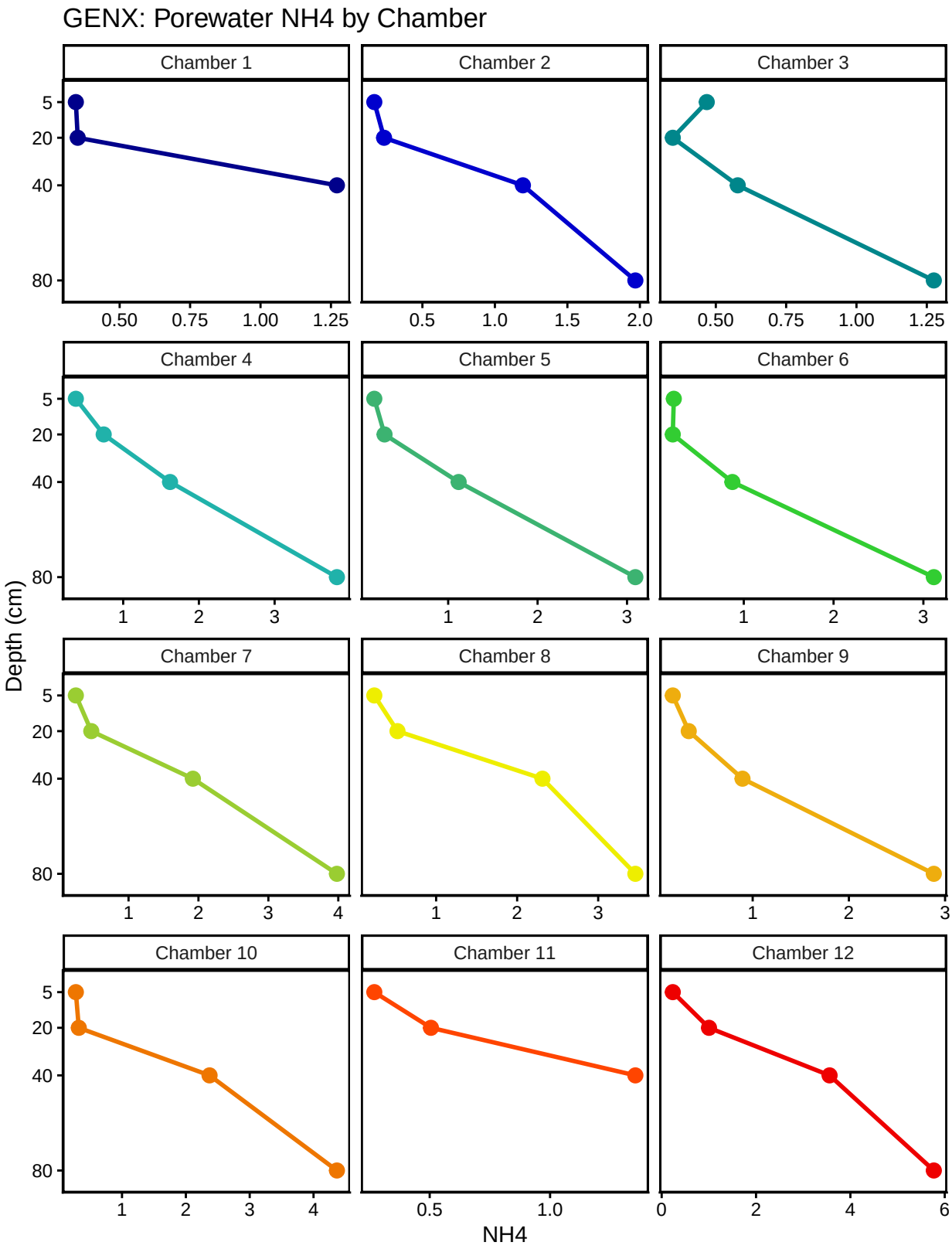
```
## Visualize Data
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_col()`).
```


GENX: Porewater NH4
Ch1
Ch2
Ch3
Ch4
Ch5
Ch6
Ch7
Ch8
Ch9
Ch10
Ch11
Ch12
5 cm
20 cm
40 cm
80 cm
0
2
4
6
NH4

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (‘geom_point()‘).

## Warning: Removed 2 rows containing missing values or values outside the scale range
## (‘geom_path()‘).
```


GENX: Porewater NH4 by Chamber
Chamber 1
Chamber 2
Chamber 3
Chamber 4
Chamber 5
Chamber 6
Chamber 7
Chamber 8
Chamber 9
Chamber 10
Chamber 11
Chamber 12
Depth (cm)
NH4

0.13 Organize

#Export Data

#end